A LEAN FORMALIZATION OF THE LONELY RUNNER CONJECTURE FOR EIGHT RUNNERS

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Abstract. For positive integer speeds $v_1, \dots, v_n$ let $\mathrm{ML}(v) = \sup_{t\in\mathbb{R}} \min_i \|v_i t\|$, where $\|x\|$ is the distance from x to the nearest integer. The lonely runner conjecture for $n+1$ runners asserts $\mathrm{ML}(v) \geq 1/(n+1)$. We describe a formalization in Lean 4 with Mathlib of the case $n = 7$ (eight runners), proved by Rosenfeld in 2025. The Lean theorem in the default build depends on the three standard axioms of Lean and on 27 axioms, each of which records a kernel check of a finite certificate that was carried out and then removed from the build for disk reasons. A second module, outside the default build, imports the 27 kernel proofs as kept object files and proves the same statement from the three standard axioms alone; the certificates are regenerated from recorded SHA-256 hashes by a published procedure. On the way the formalization proves the case $n = 6$ (seven runners, Barajas and Serra, 2008) from the standard axioms alone, and replaces the height bound of Malikiosis, Santos and Schymura used by Rosenfeld with a pigeonhole argument and the planar rounding step of Giri and Kravitz. The formalization of Barajas and Serra found errors in eleven places of the printed proof; none affects the theorem. No new mathematical result is claimed.

1. Statement and trust base

1.1. **Statement.** For $v \in \mathbb{Z}^n_{>0}$ and $t \in \mathbb{R}$ put $\mathrm{gap}_v(t) = \min_i \|v_i t\|$ and $\mathrm{ML}(v) = \sup_t \mathrm{gap}_v(t)$. The lonely runner conjecture of Wills [17] and Cusick [5] asserts

$$\mathrm{LRC}(n): \qquad \mathrm{ML}(v) \geq \frac{1}{n+1} \quad \text{for all } v \in \mathbb{Z}^n_{>0}.$$

It is known for $n \leq 6$ [3, 2], for $n = 7$ [11], for $n = 8, 9$ [12, 16], and beyond [13, 1]. In Lean, speeds are v : `Fin k` $\to \mathbb{Z}$, $\|x\|$ is `nearestIntDist` $x = \min(\{x\}, 1 - \{x\})$, and ML is the conditional supremum `sSup (Set.range (gap v))`. Repeated speeds are allowed; no primitivity is assumed.

Theorem 1.1 (Lean `LRSpectrumFull.EightRunners.eight_runners_unconditional`)**.** *For every v : `Fin 7` $\to \mathbb{Z}$ with $v_i > 0$ for all i,* $\mathrm{ML}(v) \geq 1/8$.

Theorem 1.2 (Lean `LRSpectrumFull.BarajasSerra.LRC6_barajasSerra`)**.** *For every u : `Fin 6` $\to \mathbb{Z}$ with $u_i > 0$ for all i,* $\mathrm{ML}(u) \geq 1/7$.

The module `EightRunners.lean` contains only Theorem 1.1. It imports `EightE5`, which imports the reduction, the certificate axioms and Theorem 1.2.

1.2. **Trust base.** Table 1 lists the output of `#print axioms`, run on 2026-09-27 with `lake env lean` after building the module `EightRunners`. The literal output is reproduced in the file `LEAN_MAP.md` accompanying this paper.

The 27 axioms state `lemma7Check 7 p = true` for the primes $p \in S = \{31, 37, 43, \dots, 157, 163\}$ of Rosenfeld's set, $|S| = 27$. For each p the statement was proved by the Lean kernel (`decide +kernel`) from a generated certificate, and `#print axioms` of that proof reported the three standard axioms only. The generated sources (about 1.6 GB in total) and their object files were then deleted, and the fact was reinstated as a named axiom whose docstring records the

Declaration	Axioms beyond `propext`, `Classical.choice`, `Quot.sound`
`EightRunners.eight_runners_unconditional` (Thm. 1.1)	the 27 axioms `EightCert27.lemma7Check_seven_p`*p*`_verified`
`EightCert27.hchk_all` (all 27 primes)	the same 27
`EightRunnersReal.eight_runners_unconditional_real` (same statement, outside the default build)	none
`EightRunnersReal.hchk_all_real` (all 27 primes)	none
`BarajasSerra.LRC6_barajasSerra` (Thm. 1.2)	none
`EightE5.mssGap7_holds` (rounding step)	none
`EightMSS.e5b_staircase` (pigeonhole)	none
`EightV4.lemma7Check_of_checkRootH` (checker soundness)	none

TABLE 1. Trust base. No declaration listed depends on `sorryAx`; no `native_decide` (`Lean.ofReduceBool`) is used.

SHA-256 hash of the concatenated sources. These axioms record kernel checks run by the author; the evidence that each check succeeded is the per-prime record in `code/eight/records`, and the axioms are not re-checked by a build of the present tree. All 27 certificates have been regenerated and re-checked by the kernel at least once (25 of them on 2026-09-27, the remaining two, $p = 31$ and $p = 37$, earlier), with hashes matching the recorded originals; the records in `code/eight/records` give the kernel results. Section 3.2 gives the procedure that regenerates each certificate, compares its hash, and rebuilds its kernel proof.

The module `EightRunnersReal.lean` removes the 27 axioms. It imports the modules `E8Stage.P`*p*`.Asm` produced by procedure (2) of Section 3.2 with the object files kept (about 2.6 GB in total, outside the lake package, regenerable in about 12 hours), restates `hchk_all_real` from the 27 theorems `EightV4.Cert`*p*`.lemma7Check_seven_p`*p*, and proves `eight_runners_unconditional_real` through `EightE5.eight_runners_of`, which takes the per-prime checks as a hypothesis. On 2026-09-30, compiled with `lean` against these object files, `#print axioms` reports exactly `propext`, `Classical.choice`, `Quot.sound` for both declarations. As a control, the same environment reports the 27 `EightCert27` axioms for `EightRunners.eight_runners_unconditional`. The module `EightE5` imports `EightCert27`, so the 27 axioms are present in that environment but are not used by the new theorem. This module is not built by `lake build`; it is compiled by a script in `code/eight/stage/`. For the primes $p \in \{31, 37, 43, 47, 53, 59, 61, 67, 113, 163\}$ the stored declarations were in addition replayed through a fresh kernel with `Lean.Environment.replay`, from the kept object files, against the environment of their imports; all were accepted with the three standard axioms and the statement `lemma7Check 7 p = true`. The remaining 17 primes were not replayed. This shows that the stored terms re-typecheck; it does not audit Lean, Mathlib or the checker `lemma7Check`.

2. Rosenfeld's reduction as formalized

We follow [11, Section 5] with $n = 7$. Lean names in this section are in the namespace `LRSpectrumFull.EightPlan` unless stated otherwise. A counterexample is a primitive $v \in \mathbb{Z}^7_{>0}$ with $\mathrm{ML}(v) < 1/8$ (`IsCE`).

E0 (gcd reduction) ML is invariant under division by $\gcd(v)$, so it suffices to treat primitive v (`e0_primitive`).

E4 (leave-one-out, [11, Lemma 5]) Given LRC(6), every six of the seven speeds of a counterexample are coprime; in particular at most five of them are even (`e4_leaveOneOut`, `e4_parity`).

E2 ([11, Lemmas 6, 7]) If `lemma7Check 7 p = true` then p divides some speed of every counterexample satisfying the parity condition of E4 (`e2_prime_dvd`). Here `lemma7Check` is a Boolean function defined by enumeration of admissible residue tuples modulo p.

E3 ([11, Lemma 4]) Consequently $840\prod_{p\in S} p$ divides $\prod_i v_i$ (`e3_lcm_dvd`).

NUM $28^{42} < 7^7 \cdot 840\prod_{p\in S} p$ (`numeric`), and by the inequality of arithmetic and geometric means $28^{42} < 7^7 \prod_i v_i$ implies $\sum_i v_i > 28^6$ (`e5_amgm`).

E5 Given LRC(6), a primitive $v \in \mathbb{Z}^7_{>0}$ with $\sum_i v_i > 28^6$ has $\mathrm{ML}(v) \geq 1/8$ (Section 4).

E6 LRC(6): Theorem 1.2 (Section 5).

The assembly is `EightE5.eight_runners_of`, which takes LRC(6) and the 27 prime checks as hypotheses and uses only the standard axioms; `EightE5.eight_runners_unconditional` instantiates it with Theorem 1.2 and `EightCert27.hchk_all`. A counterexample then satisfies E3, hence $\sum v_i > 28^6$ by NUM, contradicting E5.

3. The prime campaign

3.1. **The certificate checker.** Evaluating `lemma7Check 7 p` directly in the kernel is infeasible for the larger primes. We wrote a second checker, `EightCheckV4`, which walks a search tree supplied as data (a certificate) and accepts only if every leaf is closed by a counting bound or a table lookup. Its soundness theorem `EightV4.lemma7Check_of_checkRootH` (file `EightCheckSound.lean`) states that acceptance implies `lemma7Check 7 p = true`; it depends on the standard axioms only. A Rust program generates, for each prime, the certificate as Lean source split into modules; the kernel evaluates the checker on it by `decide +kernel`. The soundness proof does not depend on the generator: a wrong certificate is rejected and cannot produce a false theorem.

3.2. **Reproduction.** For each $p \in S$ a record file stores the generator command, the SHA-256 of the generator, of the checker and of the soundness file, of each generated module and of their concatenation, the toolchain (`leanprover/lean4:v4.30.0`), and the axiom output of the kernel proof. The generator, the scripts and the 27 records are in `code/eight/`. Two procedures are provided.

(1) Hash check (`stage/repro_check.sh p`): regenerate the sources and compare the concatenated hash with the record and with the axiom docstring. This takes seconds per prime and does not run Lean.

(2) Full re-check (`stage/run_prime.sh p`): regenerate, compile every module of that prime with one `lean` process per module, and require that `#print axioms` of `EightV4.Cert`p`.lemma7Check_seven_p`p reports exactly the three standard axioms.

On 2026-09-27 procedure (1), run from `code/eight/`, reproduced all 27 hashes, and every hash occurs in `EightCert27.lean`. Procedure (1) was repeated on a copy of `code/eight/` on 2026-10-18 and again reproduced all 27 hashes (1143 s; `data/repro_logs/eight_repro_all_27.txt`). Procedure (2) was not repeated for the archive. Procedure (2) was run once per prime during the campaign (2026-09-24 to 2026-09-26). Three records ($p = 31, 37, 43$) name an earlier revision of the generator source; the present generator reproduces their hashes.

4. The height bound E5

Rosenfeld obtains E5 from [9, Theorem A′], whose proof passes through zonotopes and successive minima. We did not formalize that proof. E5 is instead obtained from two statements: a pigeonhole lemma and the planar rounding step of Giri and Kravitz [7, Section 7], which reduces a speed vector close to a rational plane to LRC($n-1$). The same step is the case $r=1$ of [1, Proposition 3.4]. We claim no novelty for either.

Lemma 4.1 (Lean `EightMSS.e5b_staircase`)**.** *Let $v \in \mathbb{Z}^n_{>0}$ be primitive and $l \geq 1$ an integer with $\sum_i v_i > l^{n-1}$. There are $z \in \mathbb{Z}^n$, not proportional to v, and $\sigma \in \mathbb{Q}$ with $|lz_i - \sigma v_i| \leq 1$ for all i.*

Sketch. For each pair $e=(i,a)$ with $0 \leq a < v_i$ put $\tau_e = (2a+1)/(2v_i)$ and let $q_e \in \mathbb{Z}^n$ be the coordinatewise rounding of $\tau_e v$ with ties broken according to the index order, so that the $\sum_i v_i$ points q_e are distinct and $|q_{e,j} - \tau_e v_j| \leq 1/2$. The map $(e,c) \mapsto q_e + cv \bmod l$ on pairs with $0 \leq c < l$ has $l \sum_i v_i > l^n$ arguments, so two arguments collide. If they share e, primitivity gives $l \mid c - c'$, impossible. Otherwise $lz = q_e - q_{e'} - (c'-c)v$ defines $z \in \mathbb{Z}^n$, and $\sigma = \tau_e - \tau_{e'} - (c'-c)$ satisfies the bound. If z were proportional to v, primitivity would give $q_e - q_{e'} = sv$ with $s \in \mathbb{Z}$; the bounds $0 \leq q_{e,k}$ and $q_{e,k} < v_k$ at the index k of e (and of e') force $s=0$, contradicting $q_e \neq q_{e'}$. □

Proposition 4.2 (Lean `EightE5.mssGap7_holds`)**.** *Assume* LRC(6)*. Let $v \in \mathbb{Z}^7_{>0}$, $z \in \mathbb{Z}^7$ not proportional to v, and $\sigma \in \mathbb{Q}$ with $|28z_k - \sigma v_k| \leq 1$ for all k. Then* ML(v) $\geq 1/8$.

Proof. Choose i, j with $z_i/v_i < z_j/v_j$ adjacent among the distinct values of z_k/v_k (`EightE5.exists_adjacent_pair`). Put $c = (z_i + z_j)v - (v_i + v_j)z \in \mathbb{Z}^7$. Then $c_k = 0$ would mean $z_k/v_k = (z_i+z_j)/(v_i+v_j)$, which lies strictly between the two adjacent values; so no c_k vanishes. Also $c_j = -c_i$. Apply LRC(6) to the six positive integers $|c_k|$, $k \neq j$: there is $\tau \in \mathbb{R}$ with $\|c_k\tau\| \geq 1/7$ for $k \neq j$, and then also for $k = j$. Let $s = \sigma/28$ and $d = z - sv$, so $|d_k| \leq 1/28$. Let N be an integer nearest to $(v_i+v_j)\tau$ and $r = N - (v_i+v_j)\tau$, so $|r| \leq 1/2$. For $t = (z_i+z_j)\tau + rs$,

$$tv_k = c_k\tau + \big((v_i+v_j)\tau + r\big)z_k - rd_k = c_k\tau + Nz_k - rd_k,$$

hence $\|tv_k\| \geq 1/7 - 1/56 = 1/8$ for every k. □

With $l = 28$ and $n = 7$, Lemma 4.1 and Proposition 4.2 give E5 (`EightMSS.e5_sum_of_gap`, `EightE5.e5_sum`, and the product form `EightE5.e5_prod`). The threshold 28^6 is the one Rosenfeld uses, so no other part of the reduction changes.

Remark 4.3. An earlier version of this formalization carried E5 as an axiom transcribing the case $n=7$ of [9, Theorem A′] (`EightMSSImport.mss_theorem_A_n7`), and an attempt to formalize that theorem through its zonotope argument (`EightZonotope.lean`) stopped at [9, Propositions 3.3, 3.4]. Neither is in the closure of Theorem 1.1; both files are kept and marked as superseded.

5. Seven runners: Barajas and Serra

Lean names are in `LRSpectrumFull.BarajasSerra`. Throughout p is a prime, $N = p^{m+1}$, $\nu = \nu_p$, $q(x) = \lfloor (x \bmod N)/p^m \rfloor \bmod p$ (the arc of length N/p containing x), and Λ_j are the multiplier families of [2, Section 2].

5.1. **Prime filtering.** [2, Lemma 2] (`primeFiltering`) drives the proof: after rescaling so that the speeds sit at p-adic levels $0 \le i \le m$, a counting induction over levels produces a multiplier coprime to p that moves every speed out of the arcs $q \in \{0, p-1\}$, provided each level below the top carries few enough speeds. We formalized the digit arithmetic (q under multiplication by Λ_j, carries of $q(jd)$ for small j, reflection $q(N-d) = p-1-q(d)$), the descending induction, and the passage from $q \notin \{0, p-1\}$ to $\|td\| \ge 1/p$ for $t = \lambda/N$.

5.2. **Three and five runners: finite tables.** For $n = 2$ ($p = 3$) filtering and two short cases suffice (`barajasSerra_card2`). For $n = 4$ ($p = 5$) the configuration with three speeds at a common level and one above is not covered by filtering. We did not follow the paper's case analysis there. Each level-0 speed is replaced by a finite model (its arc, its residue mod 5, and three carry bits), and the claim that any three model elements share a good multiplier $(1+5^m k)j$ with $1 \le j \le 4$, $k < 5$ is checked by the kernel on the model (`BarajasSerraCard4Table`, eight chunks); the theorem is `barajasSerra_card4`. Both cases use the standard axioms only.

5.3. **Seven runners.** For $n = 6$, $p = 7$. After scaling the minimum level to 0, either every level below the top carries at most three speeds, and filtering applies, or one of four distributions occurs: (a) five speeds at level 0 and one above; (b) four at level 0 and two at the top; (c) four at level 0, one strictly between, one at the top; (d) one at level 0, four at an intermediate level, one at the top. The level split and dispatch are `Sec5.lrc6_nat` and `Sec5.lrc6`.

Cases (b)–(d), Section 5 of [2]. The core statement is that four level-0 units admit a common multiplier $(1 + 7^m k)j$, $j \le 5$, moving all four off the arcs 0 and 6. We prove it through a Farey model of the carries of $q(jd)$, $j \le 5$, and small kernel tables (`BarajasSerraSec5Fin`), assembled in `Sec5.sec5`. Cases (b), (c), (d) reduce to it (`Sec5.case_b0`, `case_c0` with the paper's multiplier $u'(1 + 7^{m-i_0})$, `case_d0` with filtering below level i_0).

Case (a), Sections 6 and 7 of [2]. For $m = 1$ the case is decided by a kernel table modulo 98 (`lrc6_m1_scaled`, `BarajasSerraLRC6M1Table`, 17 chunks). For $m \ge 2$ we formalized Lemmas 4, 5, 6, 7, 9, 10, 11, 12 and subsections 6.1 to 6.6 (`Sec6.sec61_witness` to `Sec6.sec66_witness`). Each subsection fixes a pattern of residue classes mod 7 of the five level-0 speeds and concludes that a multiplier coprime to 7 moves all six speeds off the arcs 0 and 6. Statements about $\mathbb{Z}_7$ alone are proved by `decide` (`BarajasSerraZ7`). The reduction of an arbitrary configuration to one of the seven patterns (sign flips, choice of base element) is `Sec6.g4`. Case (a) is `case_a`.

5.4. **Method.** Each lemma of Sections 5 and 6 of [2] was first tested in Rust with exact integer arithmetic on a transcription of the Lean definitions, in most cases exhaustively at $m = 2$ and exhaustively or by sampling at $m = 3$, together with a negative control that must fail. Only then was Lean written. The checkers cited by the errata are in `code/eight/checks/` with the commands that run them.

6. Errata to Barajas and Serra

The file `errata_barajas_serra.md` lists, for each item, the printed statement, the corrected statement, the computational evidence and the Lean declaration. In summary: Lemma 4(ii) has the wrong range in the equal-residue case; Lemma 5 is false as printed for two profiles (not used there); in Section 6 there are wrong intermediate values or signs in 6.1, 6.3, 6.5, 6.6(a) and 6.6(b), an unsatisfiable normalisation and a wrong exception pair in 6.2, an unsuitable choice of s and an untreated case $e_{45} = 0$ in 6.4, and a hypothesis of Remark 8(i) that fails where Lemma 9 uses it; in Section 5, Case 3, eq. (9) alone does not imply claim (12) (28 failures in 12544 model configurations), while (12) itself holds. None of these affects the theorem: every statement used by the argument is true, and each repair is local.

7. Scope and credits

This is a formalization of known theorems. The mathematical content of Theorem 1.1 is due to Rosenfeld [11]; that of Theorem 1.2 to Barajas and Serra [2]; the rounding step of Section 4 to Giri and Kravitz [7] (see also [1]). Rosenfeld's height bound is that of Malikiosis, Santos and Schymura [9]; the formalization avoids it, but the threshold 28^6 and the overall architecture are Rosenfeld's. Honglu Chen's repository [4] contains an earlier, independent Lean formalization of LRC(6) following Barajas and Serra; its statement was imported as an axiom, with a proved bridge to our formulation, as an interim input (`Chen7RunnerImport`). Theorem 1.1 does not depend on it. We did not re-check Chen's development, which uses a different Lean and Mathlib version.

The contributions of this paper are the Lean proofs, the certificate checker with its soundness proof, the reproduction procedure for the 27 certificates, the finite-model routes for $n = 4$ and for Section 5 of [2], and the list of errata. The formalization uses Lean 4.30.0 and Mathlib [15] pinned in the repository's `lake-manifest.json`. For background on the conjecture see [6, 14, 8, 10].

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